

A Low-Complexity Geometric PEACH Approach for 3D Localization under Elevated Planar Arrays

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Abstract—Accurate and efficient user localization is critical for Integrated Sensing and Communication (ISAC) systems, particularly in urban environments with elevated antenna arrays. This paper introduces PEACH-MUSIC (Position Estimation Algorithm using Circle and Hyperbola in MUSIC), a fundamentally novel analytical localization method. Our approach uniquely exploits an explicit geometric interpretation of the MUSIC pseudo-spectrum, a principle unexplored in existing literature, to significantly simplify the localization task. By decomposing a Uniform Rectangular Array (URA) into horizontal and vertical Uniform Linear Arrays (ULAs), the method analytically computes user positions via intersections of hyperbolic and circular spectral patterns. A Nelder-Mead NM optimization stage enhances robustness against noise. Additionally, we derive the theoretical Cramér-Rao Bound (CRB) specific to elevated antennas. Simulations demonstrate that PEACH-MUSIC achieves near-optimal accuracy, approaching the CRB while significantly reducing computational complexity, confirming its suitability for practical ISAC scenarios.

Index Terms—Direction of Arrival (DoA), Multiple Signal Classification (MUSIC), Integrated Sensing and Communication (ISAC), User Localization, Cramér-Rao Bound (CRB), Nelder-Mead Algorithm

I. INTRODUCTION

Source localization is critical in array signal processing, extensively applied in radar, sonar, wireless communications, and emerging Integrated Sensing and Communication (ISAC) systems envisioned for next-generation wireless networks [1], [2]. Among various localization techniques, the Multiple Signal Classification (MUSIC) algorithm stands out due to its high spatial resolution capabilities in estimating Direction of Arrival (DoA), applicable to near-field and far-field scenarios [3], [4]. However, limited attention has been given to practical urban deployments, where antenna arrays are elevated relative to user-stypical in base station installations atop buildings. This elevation significantly alters the MUSIC pseudo-spectrum from directional beams to localized spectral peaks [5], [6], representing a notable research gap.

With rapid developments towards 6G and ISAC frameworks, precise localization becomes essential for smart urban infrastructure, autonomous vehicles, and intelligent surveillance systems, demanding real-time and computationally efficient

techniques [7], [8]. In the context of ISAC, where sensing and communication functionalities share resources and often operate under stringent latency constraints, computationally efficient localization algorithms are paramount. A low-complexity approach not only reduces the processing load on integrated devices but also enables faster position updates, which is critical for dynamic applications and allows more resources to be allocated to communication tasks. Traditional MUSIC methods, relying on complex multi-dimensional grid searches, are computationally intensive and impractical for dynamic scenarios.

While MUSIC boasts numerous extensions, its far-field localization feature often involves computationally intensive pseudo-spectrum searches, unsuitable for real-time dynamic applications. This work addresses a less-explored alternative: leveraging elevated urban antenna arrays, where the MUSIC pseudo-spectrum is known to form localized peaks ideal for direct far-field position estimation, a scenario deserving more attention.

To overcome these challenges, the proposed Position Estimation Algorithm using Circle and Hyperbola in MUSIC (PEACH-MUSIC) method introduces a novel analytical localization by uniquely interpreting the MUSIC pseudo-spectrum. It decomposes a Uniform Rectangular Array (URA) into horizontal and vertical ULAs, which for elevated arrays distinctively yields hyperbolic and circular spectral intersections from the ULA pseudo-spectra, a novel geometric approach for this context. These intersections provide a closed-form analytical solution for 2D user coordinates, thereby avoiding computationally expensive exhaustive searches and positioning PEACH-MUSIC as an efficient, model-based MUSIC variant for such geometries.

To enhance robustness and improve 3D localization accuracy with minimal overhead, a Nelder-Mead (NM) optimization stage [9], [10] is applied to the full URA pseudo-spectrum. NM was selected for its simplicity, computational efficiency, and noise robustness in this context. The present study focuses on the single-user scenario to clearly establish this geometric principle and demonstrate its efficiency, with the extension to multi-user environments identified as a key direction for future investigation.

Additionally, we derive a theoretical Cramér-Rao Bound (CRB) tailored for elevated antenna array configurations,

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establishing a rigorous benchmark for performance validation. Numerical simulations confirm that PEACH-MUSIC achieves near-optimal accuracy close to the CRB while significantly reducing computational complexity, demonstrating its practicality for ISAC applications.

The paper is structured as follows. Section II introduces the system model, Section III details the CRB derivation, Section IV describes the enhanced MUSIC algorithm, and Section VI presents the PEACH-MUSIC estimator and its NM refinement. Numerical results are provided in Section VII, and conclusions and future research directions are summarized in Section VIII.

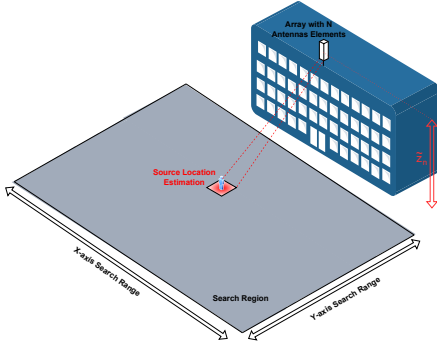


Figure 1: Representation of the elevated antenna array with the user located on the ground-level xy -plane.

II. SYSTEM MODEL

In this study, we investigate the precise localization of a single user within a defined area using an elevated antenna array. As illustrated in Fig. 1, the antennas are positioned atop a building, mimicking a realistic urban base station deployment. This elevated configuration not only improves spatial diversity, by providing a richer angular view compared to scenarios where antennas and users share the same height, but also allows the conventional MUSIC algorithm to estimate full three-dimensional coordinates. In contrast, traditional setups with co-located heights typically yield only the DoA, necessitating modifications to infer precise vertical positioning. Thus, the elevated array enables more accurate estimation of the user's horizontal (x, y) and vertical (z) coordinates, enhancing overall localization performance.

In this analysis, we consider an antenna array confined to the xz -plane, with the user's location confined to the ground-level xy -plane. Hence, the positions of the user and each antenna element are represented, respectively, by:

$$\mathbf{u} = [x \ y \ 0]^T, \quad \mathbf{p}_n = [x_n \ 0 \ z_n]^T, \quad n = 1, \dots, N, \quad (1)$$

where $\mathbf{u}$ indicates the user's position, and $\mathbf{p}_n$ specifies the location of the n -th antenna element. For elevated arrays such as the Uniform Planar Array (UPA), we define $\tilde{z}_n$ as the average height of the antenna configuration.

Given this configuration, our goal is to accurately estimate the user's location from the observed signal vector $\mathbf{r}(t)$, which is inherently linked to the steering vector $\mathbf{a}(\mathbf{u})$. The steering vector depends on the Euclidean distances between

the antenna elements and the user location, alongside path loss characteristics of the propagation channel. We consider the following radar-based signal model:

$$\mathbf{r}(t) = e^{j\psi_0} \mathbf{a}(\mathbf{u}) s(t) + \mathbf{n}(t), \quad (2)$$

where $\mathbf{r}(t)$ denotes the received signal vector at the antenna elements, $s(t)$ represents the transmitted signal from the user, and $\mathbf{n}(t)$ is the additive noise vector, with each component $n_i(t)$ assumed to be independent and identically distributed (i.i.d.) complex Gaussian noise with variance σ_n^2 . For accurate modeling of Line-of-sight (LOS) conditions at higher frequencies, our channel model incorporates the free-space path loss factor $\lambda/(4\pi)$, ensuring practical relevance and analytical clarity as validated by recent ISAC channel characterization studies [11], [12], [13]. The phase offset ψ_0 indicates the differential phase of the reflected signal relative to a reference antenna element, typically the first antenna in the array.

The steering vector $\mathbf{a}(\mathbf{u})$, capturing both directional information and path-loss attenuation, is explicitly expressed as:

$$\mathbf{a}(\mathbf{u}) = \frac{\lambda}{4\pi} \left[\frac{1}{d_1}, \frac{e^{-j\frac{2\pi}{\lambda}(d_2-d_1)}}{d_2}, \dots, \frac{e^{-j\frac{2\pi}{\lambda}(d_N-d_1)}}{d_N} \right]^T, \quad (3)$$

where d_i represents the Euclidean distance between the user and the i -th antenna element.

We assume a pure LOS scenario, appropriate for elevated antenna deployments typical in urban ISAC scenarios, especially at higher frequencies such as 15 GHz, where the LOS component typically dominates due to higher path loss in Non-Line-of-sight (NLOS) paths

III. CRAMÉR-RAO BOUND (CRB) FOR LOCALIZATION

To rigorously evaluate the theoretical accuracy limits of the proposed localization approach, we derive the CRB for estimating the user's two-dimensional position $\mathbf{u} = [x, y]^T$. The CRB is computed using the Fisher Information Matrix (FIM), quantifying the amount of information contained in the received signals relative to the position parameters.

Considering phase measurements at each antenna element, the CRB for coordinates x and y is expressed as:

$$\text{CRB}(x) = \left[J_{xx} - \frac{J_{xy}^2}{J_{yy}} \right]^{-1}, \quad \text{CRB}(y) = \left[J_{yy} - \frac{J_{xy}^2}{J_{xx}} \right]^{-1}, \quad (4)$$

with FIM elements defined as:

$$J_{xx} = P_{\text{norm}} \sum_{n=1}^N \left[\frac{\delta_n^2}{d_n^6} + \frac{4\pi^2}{\lambda^2} \left(\frac{\delta_n}{d_n^2} - \frac{\delta_1}{d_1 d_n} \right)^2 \right], \quad (5)$$

$$J_{yy} = P_{\text{norm}} y^2 \sum_{n=1}^N \left[\frac{1}{d_n^6} + \frac{4\pi^2}{\lambda^2} \left(\frac{1}{d_n^2} - \frac{1}{d_1 d_n} \right)^2 \right], \quad (6)$$

$$J_{xy} = P_{\text{norm}} y \sum_{n=1}^N \left[\frac{\delta_n}{d_n^6} + \frac{4\pi^2}{\lambda^2} \left(\left(\frac{\delta_n}{d_n^2} - \frac{\delta_1}{d_1 d_n} \right) \times \left(\frac{1}{d_n^2} - \frac{1}{d_1 d_n} \right) \right) \right], \quad (7)$$

where $\delta_n = x - x_n$, $\delta_1 = x - x_1$, and $d_n = \sqrt{(x - x_n)^2 + y^2 + z_n^2}$ represents the Euclidean distance between the user and each antenna element. The normalized power factor is defined as $P_{\text{norm}} = L \left(\frac{|s|\lambda}{\sqrt{2\pi}\sigma_n} \right)^2$, with L being the number of samples, $|s|$ the transmitted signal amplitude, and σ_n^2 the noise variance.

Notably, due to the dependence on antenna geometry and elevation, the CRB for coordinates x and y exhibits asymmetry, significantly impacting the localization accuracy according to antenna elevation and user positioning, as numerically analyzed in Section VII.

IV. ENHANCED MUSIC ALGORITHM

Conventional MUSIC implementations often rely on far-field approximations that ignore the elevation differences typical of urban deployments. In contrast, our approach explicitly incorporates the exact Euclidean distances between the user and each antenna element, ensuring higher localization accuracy. The method proceeds by first computing the sample covariance matrix

$$\hat{\mathbf{R}}_r = \frac{1}{L} \sum_{t=1}^L \mathbf{r}(t) \mathbf{r}^H(t), \quad (8)$$

then performing an eigenvalue decomposition to extract the noise subspace $\hat{\mathbf{U}}_n$. The MUSIC pseudo-spectrum is given by

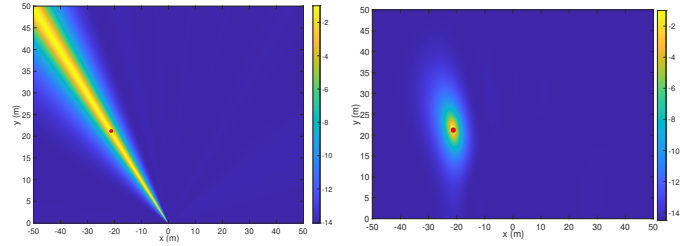
$$P_{\text{MUSIC}}(\mathbf{u}_i) = \frac{1}{\mathbf{a}^H(\mathbf{u}_i) \hat{\mathbf{U}}_n \hat{\mathbf{U}}_n^H \mathbf{a}(\mathbf{u}_i)}, \quad (9)$$

where $\mathbf{a}(\mathbf{u}_i)$ is the steering vector that depends on exact user-to-antenna distances. By locating the maximum of P_{MUSIC} , we identify the user's position with significantly improved precision relative to standard far-field assumptions.

A. Effect of Elevation on the MUSIC Pseudo-Spectrum

Elevation differences between antenna arrays and users profoundly impact the shape and localization characteristics of the MUSIC pseudo-spectrum. Unlike traditional configurations, where arrays and users are co-planar, elevated antenna arrays exhibit distinct pseudo-spectral patterns that significantly enhance localization capability and accuracy. To illustrate this, we consider a numerical scenario involving a UPA consisting of $N = 64$ antennas arranged in an 8×8 grid. Operating at a carrier frequency of 15 GHz, we evaluate the MUSIC pseudo-spectrum for a user located at 30 meters from the origin in the xy -plane with an Signal-to-Noise Ratio (SNR) of 15 dB and minimal sampling ($L = 2$).

As depicted in Fig. 2, the elevation of the antenna array transforms the MUSIC pseudo-spectrum from a conventional beam-based shape into a clearly defined peak. This effect notably improves localization performance, allowing for precise estimation of user positions rather than only DoA. Despite this enhancement, efficiently locating the pseudo-spectrum peak remains challenging due to increased computational complexity associated with searching in three-dimensional space.



(a) Antennas at user height ($\tilde{z}_n = 0$) (b) Antennas elevated ($\tilde{z}_n = 20$ m)

Figure 2: Impact of antenna array elevation on the MUSIC pseudo-spectrum for a single user. Results shown for an 8×8 planar array at $f_c = 15$ GHz, SNR = 15 dB, and sample count $L = 2$.

V. POSITION ESTIMATION ALGORITHM USING CIRCLES AND HYPERBOLAE IN PEACH-MUSIC APPROACH

The MUSIC pseudo-spectrum from a horizontal Uniform Linear Array (ULA) (antennas along the x -axis) produces a hyperbolic detection region, while a vertical ULA (antennas along the z -axis) yields a circular region in the xy -plane. The intersection of these regions, illustrated in Fig. 3, accurately determines the user's position. Considering a URA with $N_x \times N_y$ elements as two orthogonal ULAs, PEACH-MUSIC exploits this geometry for efficient localization. The vertical array provides a circle:

$$x^2 + y^2 = d_{xy}^2, \quad (10)$$

with radius d_{xy} directly estimated from the vertical ULA pseudo-spectrum.

The horizontal array defines a hyperbola based on distance differences (Δ) to antenna endpoints at symmetric positions $(-c, 0, z_a)$ and $(c, 0, z_a)$:

$$|d_1 - d_2| = \Delta, \quad (11)$$

$$\text{with } d_{1,2} = \sqrt{(x \mp c)^2 + y^2 + z_a^2}. \quad (12)$$

Solving analytically, the user's coordinates are $(x, y, 0)$, where:

$$x^2 = \frac{\Delta^2}{16c^2} [4(c^2 + d_{xy}^2 + z_a^2) - \Delta^2], \quad (13)$$

and

$$y^2 = d_{xy}^2 - x^2. \quad (14)$$

Coordinate signs are defined by MUSIC pseudo-spectrum peaks, significantly reducing computational effort compared to grid searches.

A. Efficient User Position Estimation

We propose a simplified one-dimensional scan for efficient localization. Fixing $y = 0$, the vertical and horizontal ULA spectra are sampled at N_{circle} and $N_{\text{hyper}} = 2N_{\text{circle}}$ points, respectively. Identifying the spectral peaks, the user's position is computed directly from the closed-form equations above, resolving coordinate signs using the horizontal ULA spectrum.

Algorithm 1 summarizes this efficient localization process, emphasizing its computational simplicity and practical utility.

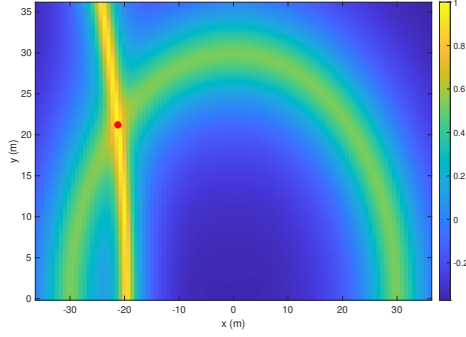


Figure 3: MUSIC pseudo spectrum combining the hyperbolic contribution from the horizontal ULA (1×8) and the circular contribution from the vertical ULA (8×1) into a single sum, for a user at 30 m in the xy -plane using an 8×8 URA at 15 GHz (SNR=15 dB, elevation $\tilde{z}_n = 50$ m).

This estimation strategy significantly reduces computational complexity while ensuring precise and efficient localization.

Algorithm 1 PEACH-MUSIC Algorithm

- 1: **Input:** Received signals from $N_x \times N_y$ URA, $x_{\max}$, N_{circle} , $N_{\text{hyper}} = 2N_{\text{circle}}$
- 2: **Output:** Estimated position $(\hat{x}, \hat{y})$
- 3: Estimate Δ from horizontal ULA pseudo spectrum line ($y = 0$)
- 4: Estimate d_{xy} from vertical ULA pseudo spectrum line ($y = 0$)
- 5: Compute user's coordinates $(x, y, 0)$ analytically:

$$x = \pm \sqrt{\frac{\Delta^2}{16c^2} [4(c^2 + d_{xy}^2 + z_a^2) - \Delta^2]}; \quad y = \sqrt{d_{xy}^2 - x^2}$$

- 6: Resolve the sign of x using horizontal ULA pseudo spectrum
- 7: **return** $(\hat{x}, \hat{y})$

VI. LOCALIZATION ESTIMATION WITH NELDER-MEAD REFINEMENT

Accurate and computationally efficient user localization is critical for practical ISAC systems. To meet these stringent requirements, we propose a two-stage localization technique using the NM optimization algorithm [9], known for its simplicity and gradient-free operation. This method iteratively refines candidate positions using simplex transformations to efficiently converge toward a local maximum of the MUSIC pseudo-spectrum.

A. PEACH-MUSIC Enhanced by Nelder-Mead Optimization

To enhance robustness, especially at lower SNR, we introduce the PEACH-MUSIC Enhanced by Nelder-Mead Optimization (PEACH-MUSIC-NM) approach. Initially, the PEACH-MUSIC algorithm provides a low-complexity, analytical estimate of the user's location by determining the intersection of hyperbolas and circles derived from the MUSIC pseudo-spectrum of orthogonal array segments. Using this initial estimate, the NM algorithm performs a refined local search to improve localization accuracy.

Two parameters significantly impact this refined search stage: the simplex side length δ_{area} and the iteration count N_{iter} .

Specifically, δ_{area} determines the side length of the initial simplex, centered around the PEACH-MUSIC estimate, defining the initial search region size. The parameter N_{iter} specifies the number of NM iterations executed, directly influencing the refinement precision. The effects of the number of iterations (N_{iter}) on the algorithms accuracy and convergence are thoroughly investigated in the Numerical Results section.

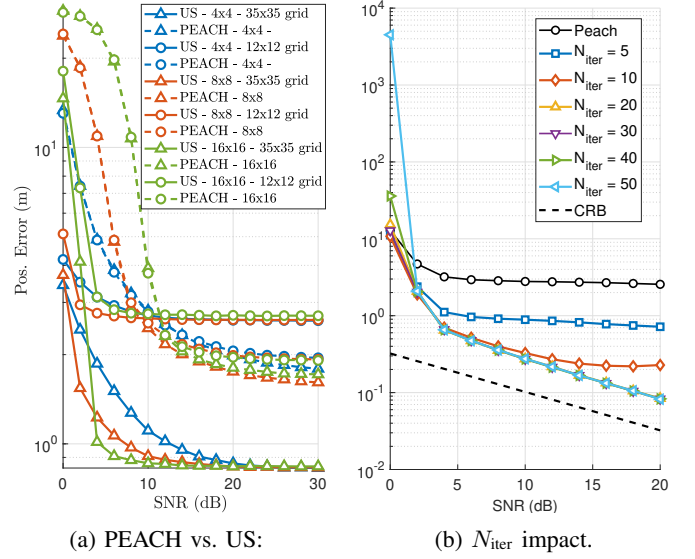


Figure 4: (a) Localization error under different array sizes and complexities. (b) N_{iter} impact on PEACH and PEACH-NM accuracy, CRB as baseline.

Table I: Simulation adopted parameter values.

Standard Parameters (Default Settings)	
Carrier Frequency	15 GHz
Antenna Config.	8×8 URA, spacing = $\lambda/2$,
Array Elevation ($\tilde{z}$)	20 m
Search Region	$x \in [-50, 50]$ m, $y \in [10, 50]$ m
User Position	$[-30/\sqrt{2}, 30/\sqrt{2}, 0]$
# Samples	$L = 100$ Samples
# ULA Vert. Search	$N_{\text{circle}} = 48$
# ULA Hor. Search	$N_{\text{hyper}} = 96$
SNR Range	SNR = $[0 : 2 : 20]$ dB
Figure-Specific Variations	
Figure (4a)	
User Position	$\mathbf{u} \sim \mathcal{U}([x_{\min}, x_{\max}] [y_{\min}, y_{\max}] 0)$
# ULA Vert. Search	$N_{\text{circle}} = [48, 408]$
# ULA Hor. Search	$N_{\text{hyper}} = [96, 816]$
# Uniform Search	$N_{\text{grid}} = [12, 35]$
Figure (4b)	
# NM Iter.	$N_{\text{iter}} = [5, 10, 20, 30, 40, 50]$
Simplex Init. Side.	$\delta_{\text{area}} = 3$ m
Monte-Carlo Simulation	
# Realizations	$\mathcal{T} = 10000$

VII. NUMERICAL RESULTS

Simulation parameters are summarized in Table I. A 15 GHz carrier frequency is employed due to its reduced wavelength, which enhances spatial resolution and aligns with

the frequency bands anticipated for future 6G urban ISAC applications. The search region spans $x \in [-50, 50]$ m and $y \in [10, 50]$ m, representing typical urban scenarios. An 8×8 URA with half-wavelength spacing, mounted at 20 m height, models standard urban infrastructure. Moreover, a sample size of $L = 100$ is chosen to maintain low complexity while ensuring adequate performance, striking a balance between computational efficiency and resolution. The value of $\mathcal{T} = 10000$ realizations ensures statistical reliability and convergence of the simulation results.

A. Coarse Estimation: PEACH-MUSIC vs. Uniform Search

We evaluate PEACH-MUSIC against Uniform Search (US) under matched complexity. Two scenarios are considered: a low-complexity scenario based on a 12×12 grid and a high-complexity scenario based on a 35×35 grid, simulated across varying array sizes. For the low-complexity case, we derive the number of circular search points using the relation $N_{\text{circ}} = \frac{N_{\text{grid}}^2}{3}$, yielding 48 points, with the hyperplane search points set to twice that value ($N_{\text{hyper}} = 96$). In the high-complexity scenario, we maintain a similar performance-complexity trade-off by selecting practical values of $N_{\text{circ}} = 408$ and $N_{\text{hyper}} = 816$.

Results in Fig. 4a show that, in the low-complexity scenario, US performance plateaus beyond 10 dB SNR due to limited grid resolution. PEACH-MUSIC significantly improves performance within this plateau, highlighting its advantage in low-complexity scenarios. In the higher-complexity scenario, US benefits considerably from the increased grid points, while PEACH-MUSIC shows marginal gains, indicating PEACH-MUSIC reaches near-optimal performance with fewer points.

It is important to note, as evident in Fig. 4a for low SNR values (e.g., below 5 dB), the performance of the PEACH-MUSIC estimator, like all MUSIC-based methods, inherently degrades. This sensitivity in the initial PEACH-MUSIC estimate can be partly attributed to its foundational step of decomposing the URA into lower-dimensional ULAs (for estimating d_{xy} and Δ). In this stage, each ULA-based estimation does not benefit from the full diversity of all N^2 URA elements, potentially leading to increased susceptibility to noise prior to any subsequent refinement stages, such as the NM optimization discussed later.

Thus, PEACH-MUSIC offers substantial advantages in scenarios with limited computational resources, whereas US can excel at higher complexities, though its computational effort may be better utilized by subsequent refinements such as NM optimization.

B. Nelder-Mead Parameter Sensitivity

In these simulations, the users position remains fixed to provide a stable baseline for variance calculation and CRB comparison. The user is positioned diagonally (as specified in Table I), representing a realistic scenario for evaluating the methods ability to localize targets in both dimensions simultaneously. Empirical simulations determined $N_{\text{iter}} = 25$ and $\delta_{\text{area}} = 3$ m as values with good performance for this scenario.

Subsequently, we conducted further experiments by varying one parameter at a time while keeping the other constant to assess the sensitivity and robustness of the algorithm.

In Fig. 4b, we evaluate the performance improvement obtained by combining the NM algorithm, considering $\delta_{\text{area}} = 3$ m, with the PEACH-MUSIC technique. It can be clearly seen that incorporating NM significantly enhances the estimation precision. Additionally, the figure illustrates how the number of iterations (N_{iter}) impacts performance. Interestingly, we observe that for values of N_{iter} greater than 30, the performance gain becomes minimal. This behavior suggests the importance of carefully calibrating the number of iterations to reduce computational complexity without substantial loss of accuracy. Furthermore, the performance gap compared to the CRB reveals an inherent limitation of the method. Nevertheless, in this scenario, the observed difference at medium to high SNR levels is in the order of centimeters, which can be considered negligible in several practical applications.

The method also demonstrated low sensitivity to the initial simplex side length (δ_{area}), as the PEACH algorithm provides a sufficiently accurate initial estimate to effectively guide the NM refinement.

The ability of the NM optimization, as used in PEACH-MUSIC-NM, to achieve lower positioning error than a computationally feasible US lies in its search mechanism. Its predefined grid resolution inherently limits a US. It will select the grid point with the maximum pseudospectrum value, potentially missing the true peak if it lies between the grid points. In contrast, NM performs an iterative simplex search directly on the continuous MUSIC pseudo-spectrum surface. Initiated by the PEACH-MUSIC estimate, NM can dynamically adapt its search steps to converge closer to the actual (off-grid) coordinates of the spectral peak, thereby often attaining a more precise localization estimate than a US constrained by a practical grid density.

C. Computational Complexity Analysis

The computational complexity differs significantly among the evaluated localization methods. In the US approach, the full URA with N^2 antennas is used. Thus, the covariance matrix has size $N^2 \times N^2$, yielding complexity $\mathcal{O}(LN^4)$, while the spectral decomposition incurs $\mathcal{O}(N^6)$. Pseudo-spectrum evaluation over a 2D grid with N_{grid}^2 points leads to $\mathcal{O}(N_{\text{grid}}^2 N^2)$, resulting in total complexity:

$$\mathcal{C}_{\text{US}} = \mathcal{O}(LN^4 + N^6 + N_{\text{grid}}^2 N^2). \quad (15)$$

For the PEACH method, only one row and one column (i.e., two ULAs of length N) are used. Therefore, the covariance and decomposition stages are each computed for $N \times N$ matrices, with costs $\mathcal{O}(LN^2)$ and $\mathcal{O}(N^3)$, respectively. The pseudo-spectrum is evaluated at $3N_{\text{circ}}$ points, each with cost $\mathcal{O}(N)$, giving:

$$\mathcal{C}_{\text{PEACH}} = \mathcal{O}(LN^2 + N^3 + 3N_{\text{circ}}N). \quad (16)$$

The PEACH-NM variant extends PEACH by refining the

estimate using N_{iter} Nelder-Mead iterations, where each iteration involves evaluating the full pseudo-spectrum using all N^2 antennas, resulting in complexity $\mathcal{O}(N_{\text{iter}}N^4)$. The total complexity is:

$$\mathcal{C}_{\text{PEACH-NM}} = \mathcal{O}(LN^2 + N^3 + 3N_{\text{circ}}N + N_{\text{iter}}N^4). \quad (17)$$

While theoretical, this analysis highlights the substantial efficiency gains of the proposed methods. Fig. 5 presents a comparative analysis of the simulation time and localization error for two distinct scenarios, both evaluated at an SNR of 20 dB, with $N_{\text{iter}} = 20$ and an area increment $\delta_{\text{area}} = 2$ m. These configurations have previously shown effective performance for NM refinement method. The first scenario, characterized by lower computational complexity, was configured with $N_{\text{cir}} = 48$, $N_{\text{hyper}} = 96$, and a 12×12 grid for US. The second scenario, representing a more complex environment, utilized $N_{\text{cir}} = 408$, $N_{\text{hyper}} = 816$, and a 35×35 grid for US.

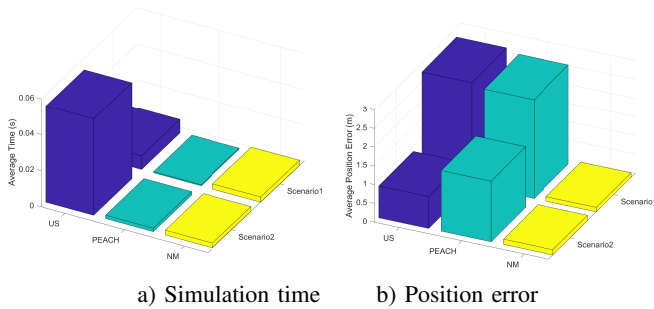


Figure 5: Simulation time and Localization error for US and PEACH-based estimators

In terms of computational efficiency, both PEACH and PEACH-NM methods significantly outperform the US approach, highlighting substantial efficiency gains. However, in scenarios with higher complexity, the US method presents a slightly lower localization error compared to PEACH alone. Despite this slight advantage, the significant increase in computational cost associated with the US method becomes evident when analyzing the simulation times.

Moreover, when examining the performance of the Nelder-Mead refinement separately, it clearly demonstrates superior efficiency, even when combined with the PEACH method, providing better localization accuracy at reduced computational costs compared to US. Thus, employing PEACH for initial estimates, followed by a targeted refinement with Nelder-Mead, proves to be a strategic approach, optimizing computational resources while maintaining high positioning accuracy, especially in scenarios of increased complexity.

VIII. CONCLUSION

This paper introduces PEACH-MUSIC. This analytical localization method leverages geometric insights from the MUSIC pseudo-spectrum for efficient and accurate user positioning in ISAC systems with elevated antenna arrays. By decomposing a URA into orthogonal ULAs, the method achieves closed-form position estimates, significantly reducing computational

complexity compared to conventional grid search techniques. The inclusion of NM optimization further enhances robustness under noisy conditions. Numerical results validated the approach against the derived CRB, confirming its near-optimal accuracy and practical viability, achieving similar performance at only a fraction of the computational cost of traditional methods. This significant reduction in computational overhead, while preserving near-optimal accuracy, also underscores the suitability of PEACH-MUSIC for ISAC systems, particularly. By minimizing the processing demands for the localization sensing task, PEACH-MUSIC can facilitate a more efficient allocation of shared resources between the integrated sensing and communication functions, thereby paving the way for more agile and responsive dual-functional wireless networks.

Future work will extend the method to multi-user scenarios, focusing on the performance and complexity tradeoff in practical configurations. Currently, the proposed PEACH-MUSIC method assumes a dominant LOS scenario. While this is reasonable for elevated antenna configurations in urban areas, practical deployments will inevitably encounter multipath and NLOS conditions.

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